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Gegeven

$$f(x) = (x^2 + 5x - 6)(x^2 - 2x + 4)$$

Methode 1 eerst uitwerken dan afleiden

$$\begin{aligned} f(x) &= (x^2 + 5x - 6)(x^2 - 2x + 4) \\ &= x^2(x^2 - 2x + 4) + 5x(x^2 - 2x + 4) - 6(x^2 - 2x + 4) \\ &= x^4 - 2x^3 + 4x^2 + 5x^3 - 10x^2 + 20x - 6x^2 + 12x - 24 \\ &= x^4 + 3x^3 - 12x^2 + 32x - 24 \end{aligned}$$

Afgeleide nemen

$$f'(x) = 4x^3 + 9x^2 - 24x + 32$$

Methode 2 productregel

$$u = x^2 + 5x - 6 \quad v = x^2 - 2x + 4$$

$$f'(x) = u'v + uv'$$

$$u' = 2x + 5 \quad v' = 2x - 2$$

$$\begin{aligned} f'(x) &= (2x + 5)(x^2 - 2x + 4) + (x^2 + 5x - 6)(2x - 2) \\ &= 2x^3 - 4x^2 + 8x + 5x^2 - 10x + 20 \\ &\quad + 2x^3 + 10x^2 - 12x - 2x^2 - 10x + 12 \\ &= 4x^3 + 9x^2 - 24x + 32 \end{aligned}$$

Beide methodes geven dezelfde afgeleide

$$f'(x) = 4x^3 + 9x^2 - 24x + 32$$

References